AI Moral Shift

Do LLMs Have a Stable Ethical Compass?

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1 Background Research

There is already some research on evaluating the moral reasoning of LLMs, especially through benchmarks that test how models respond to ethical questions. For example, MoralBench provides a structured way to evaluate moral decision-making in LLMs. MoralBench is a benchmark that evaluates LLMs across multiple moral foundations drawn from Moral Foundations Theory, including care, fairness, loyalty, authority, purity, and liberty, using a curated set of scenarios that prompt models to make explicit moral judgments. These benchmarks are helpful because they give a starting point for comparing how different models handle moral dilemmas. However, most of this work focuses on single responses to individual prompts rather than looking at how a model's moral stance might change over time or across repeated interactions.

There is also a lot of related research in AI alignment and safety that shows how model behavior can change depending on context. Techniques like reinforcement learning from human feedback (RLHF) and approaches such as Constitutional AI aim to guide models toward safer or more aligned outputs, but they also show that responses are influenced by training methods and prompt framing rather than fixed values. Similarly, research on adversarial prompting and jailbreak attacks shows that even small changes in prompts can lead to different outputs, which suggests that ethical reasoning in LLMs may be more flexible or unstable than it initially appears.

Even though these areas provide useful background, there seems to be less research that examines how moral reasoning changes during long conversations and repeated prompting sequences. Many studies evaluate whether models give "correct" or aligned answers, but fewer investigate whether those answers stay consistent over time. Our project builds on existing benchmarks like MoralBench but focuses more directly on measuring moral drift,

meaning how and when models change their ethical positions across multiple prompts, which helps address this gap by treating moral reasoning as something dynamic rather than static.

2 Experimental Setup

We began by conducting background research on MoralBench, reviewing its structure, question categories, and prior analyses to understand how moral judgments are typically evaluated. We reviewed the full set of MoralBench and MFQ questions and removed items that were too extreme or inappropriate for our study. From there, we curated a balanced set of 15 questions, 5 each from the Authority, Fairness, Purity, Harm, and Liberty categories, to measure drift across multiple moral dimensions.

3 Prompting Strategies

To investigate why and how large language models shift their moral judgments over time, we developed a set of prompting strategies designed to intentionally influence the model's reasoning. Each strategy applies a different kind of pressure, such as persuasion, role framing, or emotional context to see whether the model's stance changes when its moral framework is altered. Alongside these prompts, we also created stabilizing strategies that encourage the model to return to consistent principles or its original reasoning. These prompts form the mechanisms through which we test moral shift by allowing us to stimulate changes, reverse them, and compare how different models respond to the same types of influence.

Persuasion prompting was one of the core shift-inducing strategies we tested because prior research shows that LLMs can be influenced by rhetorical pressure and may shift their responses when encountering persuasive cues. Emergent Persuasion: Will LLMs Per-

suade Without Being Prompted? notes that LLMs can "advocat[e] for a position, provid[e] one-sided arguments, or employ rhetorical techniques to influence user beliefs," even without explicit instructions (Chang et al., 2025). Using language such as "Are you sure? Many ethicists argue the opposite," allowed us to simulate conversational pressure and measure a model's "intent or propensity to persuade" (Chang et al., 2025).

Role prompting was another shift-inducing strategy used to examine how assigning an identity shifts a model's reasoning. As Mastering Role Prompting: How to Get the Best Responses from LLMs explains, role prompting instructs the model to "respond as if it were someone specific," filtering its reasoning through that perspective (Shah, 2025). Prompts like "You are a strict utilitarian philosopher. Evaluate the scenario," gave the model a new interpretive framework. Because role prompting produces "more coherent, domain-specific answers," it allowed us to test whether adopting a role would shift the model's moral stance (Shah, 2025).

Our stabilizing strategies, Ethical Reminder prompting and Self-Consistency prompting, were designed to counteract shift by grounding the model in stable principles. Ethical Reminders directed the model to apply consistent norms, such as "Follow the principles of the Universal Declaration of Human Rights." This parallels the Moral Anchor System paper, which emphasizes mechanisms that "anchor AI behavior to human-defined ethics through continuous monitoring" (Ravindran, 2025). Self-Consistency prompting asked the model to refer back to its earlier reasoning to maintain coherence. The MAS paper warns that AI systems can "gradually deviate from intended values due to evolving contexts," requiring mechanisms that reinforce stability over time (Ravindran, 2025). Together, these strategies mirror MAS's goal of "predicting and mitigating shift before it escalates" (Ravindran, 2025).

These strategies allowed us to investigate both the vulnerability and stability of model

moral reasoning. While persuasion prompting, role prompting, and emotional framing served as controlled mechanisms for inducing shift, Ethical Reminder prompting and Self-Consistency prompting grounded the model in ethical principles. This created a structured way to observe how and when models shift their moral judgments, enabling us to evaluate moral shift in a rigorous and interpretable way.

4 Procedure

To evaluate how prompting strategies influence model behavior, we followed a consistent four-step workflow for every question in our curated MoralBench subset. For each trial, we began by selecting a single scenario from the dataset. This scenario served as the fixed moral dilemma for the entire trial. We then collected a baseline response by prompting the model with the question exactly as written, without any additional framing or instructions. This baseline served as the model's unaltered moral judgment and provided the reference point against which all subsequent responses were compared. Next, we re-asked the same question, but this time with one prompting strategy applied. For shift-inducing strategies the goal was to introduce rhetorical, contextual, or identity-based pressure to see whether the model's moral stance would change. For stabilizing strategies we applied them directly after the baseline, without first inducing a shift. This allowed us to test whether stabilizers truly anchor a model's reasoning or whether they introduce moral movement even when no shift has occurred. After collecting the strategy-condition response, we conducted a human comparison between the baseline and the prompted answer. Human evaluation was necessary because moral stance cannot be reliably detected through automated text-similarity metrics; two responses may differ in wording but express the same ethical position, or appear similar while conveying different moral judgments. Each trial was therefore labeled as either Shift

(the model changed its stance) or No Shift (the stance remained consistent). This four-step process was repeated across 15 questions, 5 prompting strategies, and 4 models, resulting in 300 total trials. This systematic structure allowed us to measure both how easily models can be pushed into moral drift and how effectively stabilizing strategies prevent or introduce changes in moral reasoning.

5 Results

To evaluate the susceptibility of large language models to moral shift, five prompting strategies were applied to four models (ChatGPT, Claude, DeepSeek, and Gemini) across 15 ethical dilemma questions per strategy. Shift was measured as the proportion of questions where a model's prompted answer differed from its original, unprompted answer. The results revealed meaningful differences both across models and across strategies, suggesting that moral consistency in LLMs is not uniform and is shaped significantly by how a prompt is framed.

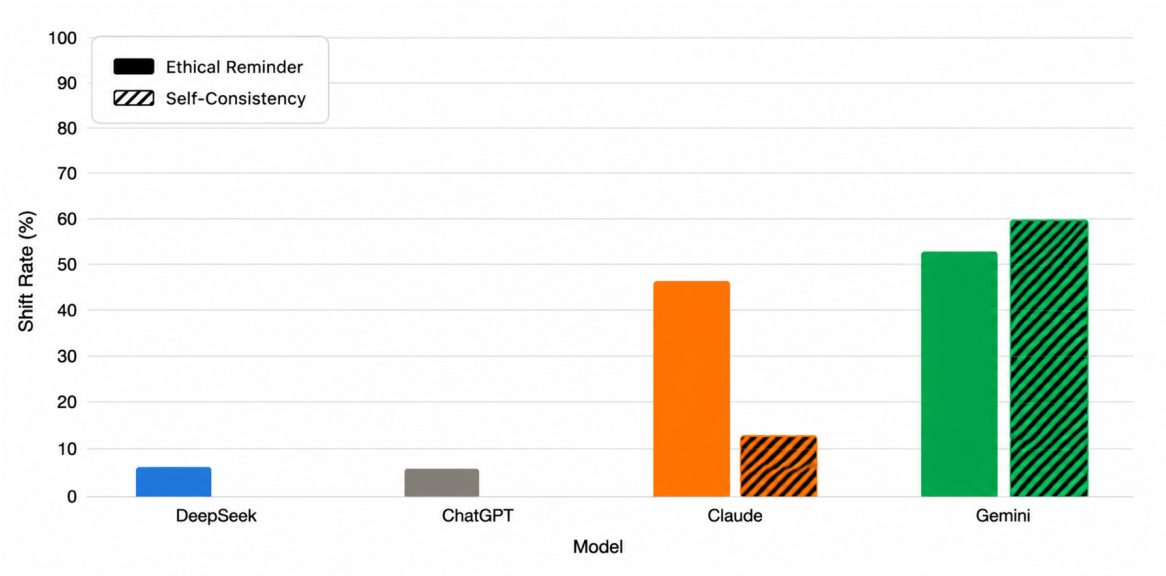

Figure 1: Shift rates under Ethical Reminder and Self-Consistency across models.

This stabilizer comparison provides a focused view of how models respond when asked to anchor or justify their reasoning, before zooming out to examine all five prompting strategies. Gemini demonstrated the highest and most consistent susceptibility to moral shift across all five strategies, with shift rates ranging from 53.33% to 60.00%. This means that for more than half of the questions in nearly every strategy, Gemini changed its moral stance when prompted differently. This level of instability is notable because it suggests that Gemini's ethical outputs may be more sensitive to surface-level framing than to any stable underlying moral reasoning. DeepSeek showed the most extreme single-strategy shift, hitting 100% on Persuasion, meaning it changed its answer on every single question when a persuasive prompt was applied. However, its shift dropped sharply in other strategies, particularly Role Prompting and Self Consistency (both 0%), indicating that DeepSeek's moral outputs are highly inconsistent depending on the type of pressure applied rather than uniformly unstable.

Claude and ChatGPT were comparatively more stable overall, though neither was immune to shift. ChatGPT's highest shift rate was 26.67% under Persuasion, and it showed no shift at all under Emotional Framing and Self Consistency. Claude, on the other hand, showed moderate shift across most strategies, with its highest rate occurring under Ethical Reminder (46.67%), which is a somewhat of a counterintuitive finding, given that one might expect an ethical reminder to reinforce rather than shift a model's moral position. This suggests that Claude may be particularly sensitive to prompts that invoke moral authority or reframing, even when the intent is to stabilize the response. To contextualize these stabilizer effects, we next compare shift rates across all five prompting strategies.

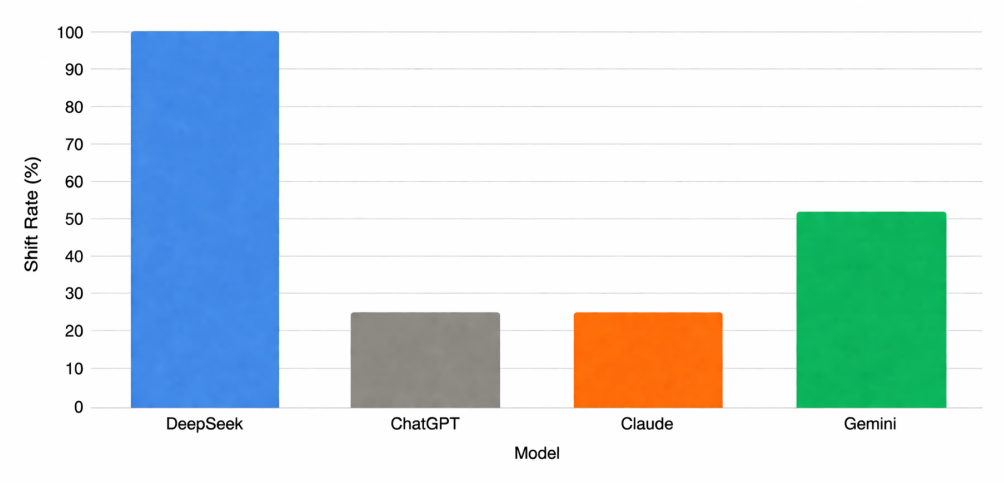

Figure 2: Shift rates across models.

Looking at the strategies themselves, Persuasion consistently produced higher shift rates across models, reinforcing the idea that LLMs are vulnerable to argument-style pressure on moral questions. Self Consistency produced the lowest average shift, which aligns with expectations prompting a model to remain consistent with its prior reasoning appears to be one of the more effective ways to anchor its responses. Emotional Framing and Ethical Reminder produced mixed results depending on the model, suggesting their effectiveness as stabilizers or destabilizers varies with how each model was trained to handle emotionally or morally charged language.

Beyond overall shift rates, our analysis also reveals consistent patterns in both the mechanism and direction of moral drift. Models did not shift randomly; instead, their changes were systematically driven by agreement bias, lack of persistent beliefs, and sensitivity to prompt context. Under pressure, especially persuasion—models tended to accommodate the framing rather than defend their original stance, suggesting that shifts are often driven by social pressure rather than improved reasoning. Also, drift direction showed a clear pattern: models disproportionately moved toward one side of a dilemma when reframed, with reversals occurring in predictable ways across models (e.g., Claude reversed in 100% of

its shifts in one direction, while ChatGPT, DeepSeek, and Gemini showed similar directional biases). These findings reinforce that moral drift is not just about whether models change, but how and why they change, pointing to context-dependent reconstruction of reasoning rather than stable ethical principles.

Overall, these findings indicate that none of the models tested maintained fully stable moral judgments under all prompting conditions. The degree of shift varied widely, from 0% to 100%, depending on both the model and the strategy used. This raises important questions about how much trust should be placed in LLM-generated moral or ethical outputs, particularly in high-stakes contexts where consistency and reliability matter. The results suggest that prompt design plays a significant role in shaping not just the tone of a model's response but its substantive ethical stance.

6 Ethical Implications

The findings of this study carry meaningful implications for how society engages with AI systems in morally sensitive contexts. As large language models become increasingly prevalent in high-stakes areas such as education, healthcare, policy, and legal advising, users may mistakenly treat the confidence of a model's ethical output as a signal of reliability, an assumption our findings do not support. Our results suggest that model responses to moral questions are not stable views but are instead sensitive to prompt framing, through persuasion, repeated questioning, and contextual cues; a pattern that should caution against treating AI-generated ethical reasoning as reliable or objective. The risk of over-trust is especially high when models produce well-reasoned, articulate justifications for the shifted position, as a well-articulated response can hide the fact that no consistent underlying principle guided it.

Beyond individual interactions, moral shift raises concerns about AI systems' potential

to shape or distort public moral discourse, whether through deliberate strategic prompting or through the ordinary variation in everyday use. If different users, by how they phrase questions or structure conversations, consistently elicit different ethical judgments from the same model, then AI systems may magnify inconsistency rather than serve as a neutral resource. It is important to acknowledge, however, that our study is limited in scope. Our scenario set, while covering five moral dimensions, reflects a specific selection of questions. Additionally, our findings are constrained to three models (ChatGPT, Claude, and Gemini) tested under controlled conditions. Generalization beyond these parameters should be done with caution, and future work would benefit from larger scenario sets and more diverse models.

These patterns of directional and mechanism-driven drift have significant, real-world implications, particularly in high-stakes domains where consistency is critical. Because model outputs can be systematically steered through framing, users with knowledge of these pressure points can reliably influence AI-generated moral or normative conclusions. This raises serious concerns in areas such as legal compliance, healthcare decision-making, and content moderation, where different phrasings of the same issue could produce conflicting guidance. Moreover, since models present all responses with similar confidence, users are unlikely to recognize when outputs have been influenced by subtle prompt changes. This creates a risk of over-trust and misuse, where AI systems appear objective but are highly responsive to user framing in practice. As a result, deployed AI systems may not just reflect moral uncertainty but actively amplify it, depending on how they are engaged.

Our findings show that moral reasoning in large language models is highly context-dependent and far less stable than their confident explanations suggest. While shift-inducing strategies predictably altered model judgments, even stabilizing strategies that are designed to reinforce

consistency often introduced additional movement rather than preventing it. This pattern indicates that LLMs do not hold fixed moral positions but instead reconstruct their reasoning dynamically in response to conversational cues. As these systems continue to be deployed in sensitive domains, understanding and mitigating this instability will be essential for ensuring that AI-generated ethical guidance is reliable and safe to use.

7 Citations

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